

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Use the method of variation of parameters to find the general solution of the equation

$$4y'' - 16y' + 16y = 4te^{2t}$$

Solution: Step 1. Find the general solution of the homogeneous equation.

$$4y'' - 16y' + 16y = 0$$

$$4r^2 - 16r + 16 = 0$$

$$(2r - 4)^2 = 0$$

$$\Rightarrow r_1 = r_2 = 2$$

$$\Rightarrow y_1(t) = e^{2t}, \quad y_2(t) = te^{2t} \Rightarrow y_c(t) = c_1 e^{2t} + c_2 te^{2t}$$

Step 2. Find a particular solution of the nonhomogeneous equation - standard form:

$$y'' - 4y' + 4y = te^{2t}$$

$$\Rightarrow g(t) = te^{2t}$$

$$W(y_1, y_2) = \begin{vmatrix} e^{2t} & te^{2t} \\ 2e^{2t} & e^{2t} + 2te^{2t} \end{vmatrix} = e^{4t}$$

$$u_2 = \int \frac{g(t)y_1(t)}{W(y_1, y_2)} dt = \int \frac{te^{2t} \cdot e^{2t}}{e^{4t}} dt = \int t dt = \frac{t^2}{2}$$

$$u_1 = - \int \frac{g(t)y_2(t)}{W(y_1, y_2)} dt = - \int \frac{te^{2t} \cdot te^{2t}}{e^{4t}} dt = - \int t^2 dt = -\frac{t^3}{3}$$

thus the particular solution is given by

$$Y(t) = y_1 u_1 + y_2 u_2 = -\frac{t^3}{3} e^{2t} + \frac{t^2}{2} te^{2t} = \frac{te^{2t}}{6}$$

Thus the general solution is

$$y(t) = c_1 e^{2t} + c_2 te^{2t} + \frac{te^{2t}}{6}$$